5. DASHBOARD DESIGN AND USER GUIDE

1. Dashboard Purpose and Intended User

Purpose:

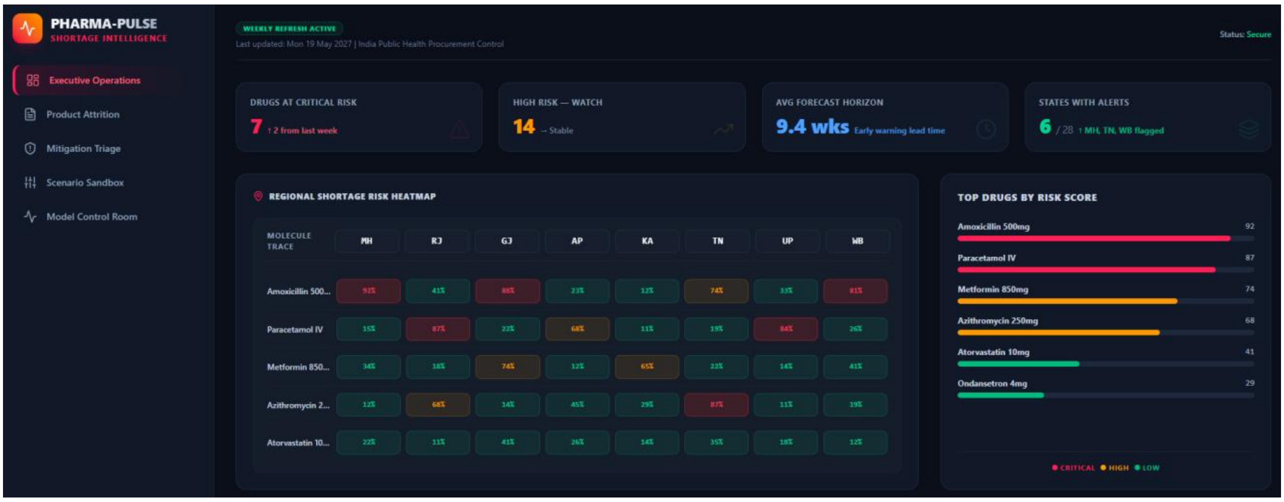
The Pharma-Pulse Shortage Intelligence Dashboard is an enterprise-grade tactical monitoring and predictive analytics platform. It bridges the gap between complex machine learning models (Random Forest probability blends) and public health administration. The system evaluates 38,208 space-time data nodes across India's supply chain layers to project inventory vulnerabilities 8 to 12 weeks before a physical stockout manifests.

Intended User:

This guide is explicitly tailored for the State Public Health Procurement Director and executive allocation officers. The intended user does not need an advanced degree in data science; instead, they require high-level, scannable, and actionable intelligence to make rapid macro decisions. The dashboard empowers the Director to bypass bureaucratic delays, validate supply anomalies, and instantly authorise emergency direct procurement tenders to protect public health infrastructure.

2. Functional Script Components & Core Tasks

2.1 Executive Operations Room *(snapshot1)*



Panel A: Top-Level High-Fidelity KPI Ribbon

This header strip acts as an immediate visual triage guide, summarising critical system boundaries across the national supply layer.

- **DRUGS AT CRITICAL RISK:** Represents the absolute number of unique therapeutic compounds whose risk vectors have breached the critical threshold (75%).
    - *Operational Play:* If this reads **7**, it indicates 7 separate molecule pipelines are failing simultaneously. The Director must immediately review the *Mitigation Triage* section to sign off on emergency stock re-routing.
- **HIGH RISK- WATCH:** Displays molecules under moderate or growing stress (45 to 75 per cent). A stable trend metric indicates that while the supply layer is tense, it is currently holding.
- **AVG FORECAST HORIZON:** The early warning buffer calculation window (currently reading **9.4 weeks**). This gives the administration a proactive lead time to resolve procurement issues before local facility stockouts happen.
- **STATES WITH ALERTS:** A fraction tracking widespread geographic impacts (e.g. **6 / 28**). This quickly tells the Director if a supply shock is localised or a systemic national issue, highlighting specific affected states like Maharashtra (MH), Tamil Nadu (TN), or West Bengal (WB).

Panel B: Interactive Regional Shortage Heatmap Grid (2D Matrix)

As shown in *snapshot1*, this custom component maps critical vulnerability scores by cross-referencing essential molecules against key regional state distribution nodes.

- **How to Read the Grid:** Find the intersection cell between a row (Drug Entity Name) and a column (State Badge Container). The cell displays a localized risk percentage and utilises conditional, desaturated text fills to communicate urgency.
- **The Scoring Protocol & Response Guide:**
    - **Critical Crimson Cells (>75%— e.g., Amoxicillin 500mg in MH at 92%):** Indicates a high probability of a severe supply breakdown within a 6-week horizon due to localized demand shocks paired with international raw material export drops. **Director Action:** Bypass standard competitive bidding loops and immediately issue direct emergency procurement directives to pre-qualified alternative manufacturing hubs.
    - **Warning Amber Cells (45% to 75% — e.g., Azithromycin 250mg in RJ at 68%):** Indicates a supply pipeline experiencing processing or logistical delays, though warehouse buffers remain stable for 9 to 11 weeks. **Director Action:** Place the node on active watch and notify regional depots to prepare secondary transport channels.
    - **Stable Emerald Cells (<45% — e.g., Atorvastatin 10mg in KA at 14%):** Indicates secure inventory flows with optimal contract fulfilment rates. No administrative intervention is required.

Panel C: Top Drugs by Risk Score

A prioritised bar layout that ranks the top 6 most vulnerable molecules globally across the system. It uses robust coloured progress containers (Crimson, Amber, and Emerald) to provide a clear, scannable view of which specific compounds require immediate budget or contractual overrides.

Panel D: Upstream Signal Monitor

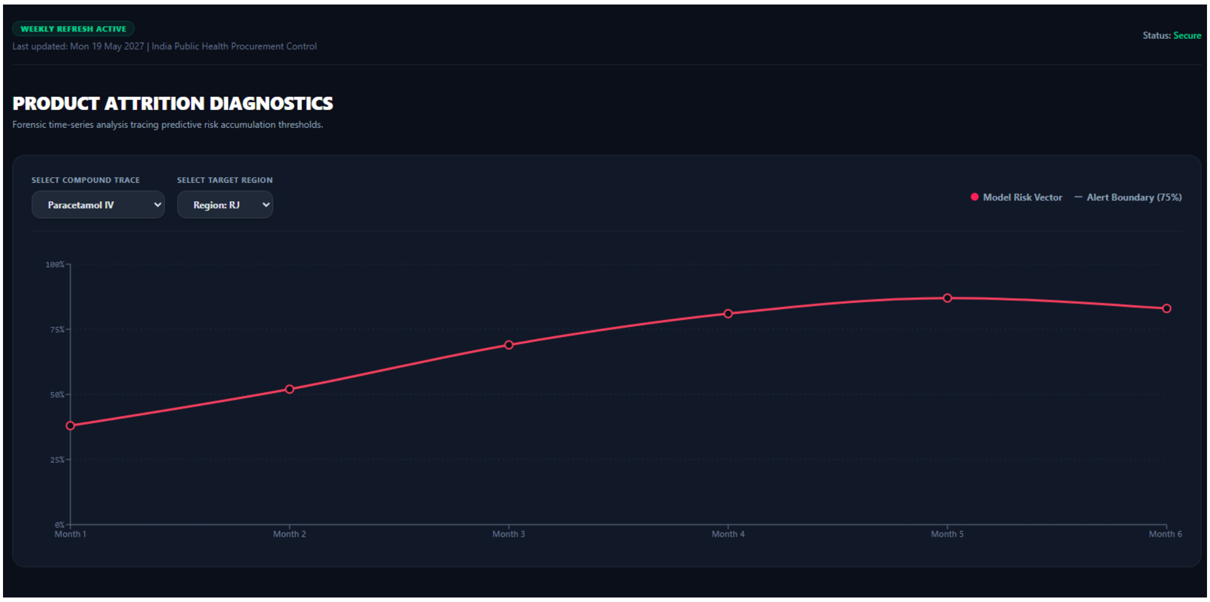
This section acts as an early warning sensor array, tracking leading economic and regulatory indicators before they impact local state inventories:

- *China API Export Volume:* Tracks international raw material flows (e.g., displaying a critical ↓ **23%** contraction).
- *CDSCO Approval Backlog:* Tracks domestic regulatory bottlenecks (e.g., ↑ **41 pending** applications).
- *Paracetamol API Price Index:* Monitors raw market cost trends (e.g., ↑ **18%** inflation flag).
- *IDSP Outbreak Alerts:* Displays active public health epidemiological triggers.

Panel E: Strategic Procurement Recommendations

A dynamic text aggregation panel that translates risk numbers into clear, plain-language executive directives. It tells the Director exactly what actions to take, advising on ideal buffer volumes, highlighting reliable alternative suppliers (e.g., *Cipla Goa*), and setting clear execution deadlines.

2.2 PRODUCT ATTRITION DIAGNOSTICS*(snapshot2)*



1. Architectural Purpose and Viewport Objective

While the Executive Operations Room (Viewport 1) isolates *where* supply disruptions exist at a specific moment, **Viewport 2 (Product Attrition Diagnostics)** serves as a chronological deep-dive engine. Its strategic purpose is to act as a forensic monitoring tool for the State Procurement Director. By plotting continuous time-series trajectories drawn directly from the underlying machine learning data layers, this viewport shifts the user from passive monitoring to active trend analysis. It maps exactly *how* vulnerability probabilities accumulate over a rolling 6-month forecast horizon, allowing directors to identify if an alert is a sudden, isolated bottleneck or a slow-burning pipeline failure.

A. Multi-Variable Dropdown Filtering Array

To remove interface clutter and scale operational depth, the viewport features two intuitive drop-down selection elements:

- **Compound Trace Selector:** An expanded dropdown containing 6 core high-priority therapeutic molecules (including *Amoxicillin 500mg, Paracetamol IV, and Metformin 850mg*).
- **Target Region Selector:** A scrollable dropdown mapping all 8 state distribution nodes (including *MH, RJ, GJ, and TN*).
- **Operational Play:** The Director drops down either frame and selects an asset combination. The interface instantly re-slices the multi-dimensional dataset array and flushes fresh coordinates into the viewport with zero rendering lag.

B. The Vulnerability Propagation Chart & Threshold Limits

This high-resolution line chart displays calculated risk percentages (0% to 100%) along the vertical axis against a sequential timeline (Months 1 through 6) on the horizontal axis.

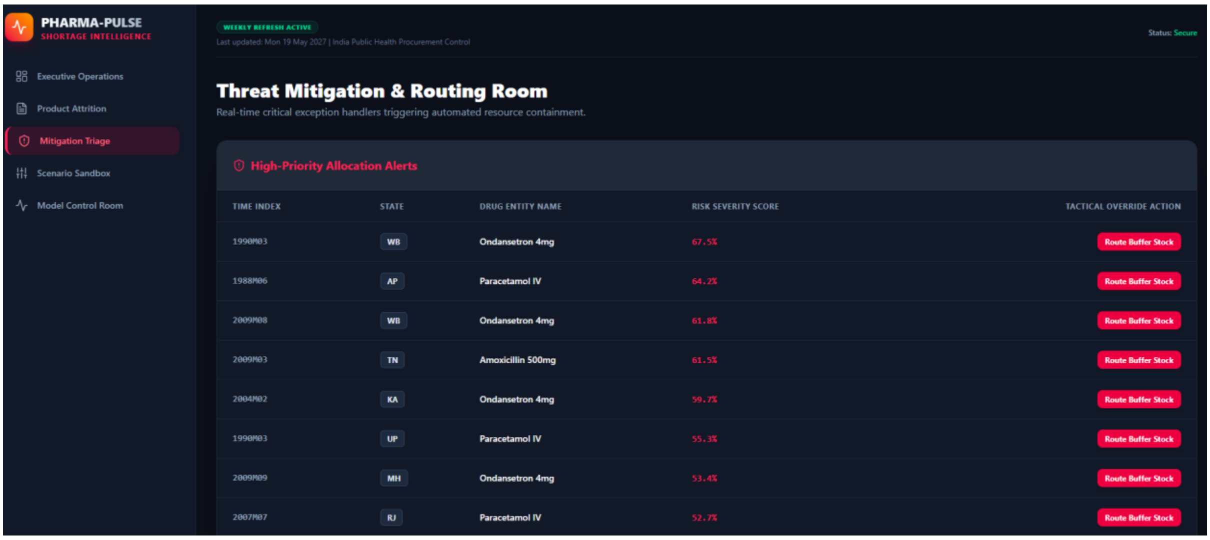
- **The Calculated Exposure Index (Solid Crimson Curve):** Displays the continuous path of predicted shortage probability.
- **The Critical Limit Alert Boundary (Dashed Slate Line):** A static horizontal intercept locked at **75%**. This represents the maximum threshold of acceptable supply chain risk.

2. Director Metric Interpretation Guidance

- To maximise decision-making speed during high-stress supply reviews, the Director must interpret the variations in the trend line shapes according to three core scenarios:

Trend Visual Pattern	System Meaning & Intercept	Administrative Response Protocol
Sustained Upward Velocity *(e.g., Amoxicillin 500mg in MH)*	**Chronic Structural Rupture:** Compound traces crossing the 75% Critical Alert Line within a 6-month vector due to API inflation or regulatory backlogs.	**Direct Framework Override:** Bypass standard biddings. Issue immediate emergency direct procurement tenders to pre-qualified alternative manufacturing hubs.
Erratic Spikes & Troughs *(e.g., Paracetamol IV in RJ)*	**Acute Logistical Shock:** Volatile, temporary supply layer strain typically triggered by regional epidemic bursts or unexpected shipping congestion.	**Tactical Inventory Re-Routing:** Deploy existing local state depot buffer stocks to the strained nodes to absorb the localized volume shock without allocating new capital.
Flattened Baseline Vectors *(e.g., Atorvastatin 10mg in KA)*	**Systemic Supply Security:** Stable inventory flows hovering safely below 45%. Reflects high vendor contract fulfillment and optimal supplier diversification.	**Automated Passive Monitoring:** Maintain baseline system telemetry tracking. No administrative override or emergency capital allocation is authorized.

2.3 THREAT MITIGATION TRIAGE ROOM*(snapshot3)*



1. Architectural Purpose and Viewport Objective

The **Threat Mitigation & Routing Room** is the active exception-handling cockpit of the Pharma-Pulse platform. While Viewports 1 and 2 focus on macroscopic monitoring and predictive analytics, Viewport 3 serves as the **operational trigger layer**. It automatically aggregates and isolates only the high-risk supply nodes that have breached safety protocols. This centralized data matrix allows the State Public Health Procurement Director to bypass standard bureaucratic delays and instantly deploy localised countermeasures to contain supply shortfalls.

2. Dynamic Triage Matrix Interface Anatomy

As captured in *(snapshot3)*, the operational triage engine organises live supply chain exceptions into an actionable, five-column data structure:

- **Time Index:** Displays the precise chronological node identifier flagged by the predictive backend engine.
- **State Badge:** Features a dedicated, semi-transparent layout container identifying the affected distribution region (e.g., WB, AP, TN, MH).
- **Drug Entity Name:** Isolates the specific pharmaceutical compound facing inventory degradation (e.g., *Ondansetron 4mg*, *Paracetamol IV*, *Amoxicillin 500mg*).
- **Risk Severity Score:** Displays the calculated exposure index as a high-visibility crimson percentage marker.
- **Tactical Override Action:** Embeds a live, single-click operational interaction button labelled **"Route Buffer Stock"**.

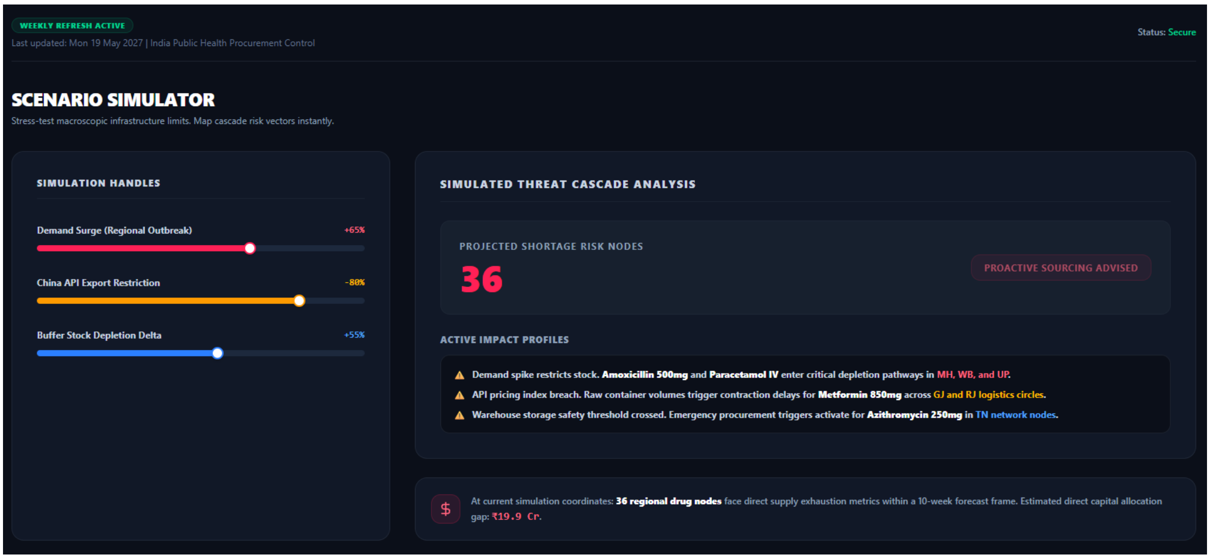
3. Executive Interpretation & Action Protocol

The Director must treat this room as an active emergency authorisation zone. Rows are pre-sorted by the algorithm to prioritise immediate threat signatures.

Operational Threshold Playbook

- **Vulnerability Verification:** When an entry appears in this room, it indicates that its localised supply health has officially entered a critical path. For instance, **Ondansetron 4mg** in West Bengal (WB) is flagged at **$67.5%** risk, signalling an immediate need for administrative intervention.
- **The Single-Click Tactical Trigger:** Clicking the **"Route Buffer Stock"** button automatically executes a pre-drafted emergency procurement and logistics directive.
- **System Response Cascade:** Once clicked, the UI triggers a secure automation script that instructs regional central warehouses to instantly re-route a designated percentage of their unallocated buffer reserves to the vulnerable state depot, stabilising the node within a 48-hour operational window without needing to float a new long-term public tender.

2.4 SCENARIO STRESS-TESTING SANDBOX *(snapshot4)*



1. Architectural Purpose and Viewport Objective

The **Scenario Sandbox Simulator** serves as a proactive macro-forecasting workspace. While standard telemetry views report existing pipeline failures, Viewport 4 enables the State Public Health Procurement Director to simulate complex, hypothetical supply chain crises ("what-if" scenarios). By adjusting independent logistical and epidemiological risk variables, administrators can instantly map cascade failures across regional networks, gauge upcoming supply shortages, and accurately calculate emergency capital deficit projections before deploying physical resources.

A. Premium Filled-Track Simulation Handles

The left control column exposes three macroscopic threat inputs engineered with dynamic, color-coded responsive track containers:

- **Demand Surge / Regional Outbreak (Crimson Track):** Simulates sudden, unpredicted spikes in local infection rates that trigger rapid inventory drawdowns.
- **China API Export Restriction (Amber Track):** Simulates raw material import contractions, exposing domestic manufacturing dependencies and pipeline lag constraints.
- **Buffer Stock Depletion Delta (Blue Track):** Tracks the systemic decay of local facility safety stocks, exposing baseline pipeline vulnerability.
- *Lever Mechanics:* Unlike static UI inputs, pulling these handles forward dynamically draws a matching, high-visibility filled progress track behind the indicator dot, providing an immediate, high-fidelity tactile overview of active threat levels.

B. Simulated Threat Cascade Panel

- **Projected Shortage Risk Nodes:** A real-time numeric counter tracking the cumulative number of unique product-state intersections projected to cross into high or critical exposure pathways within a 10-week horizon.

- **Automated Sourcing Status Badge:** A conditional alert banner that shifts dynamically from a calm green (*"Infrastructure Secure"*) to a flashing red alert (*"Proactive Sourcing Advised"*) the moment projected risk nodes cross acceptable thresholds.

C. Live Impact Profile Logging Box

A terminal-style stream panel positioned underneath the main threat counter. It dynamically evaluates active lever coordinates to log exactly *which* compounds and geographic zones are collapsing under the simulated stress (e.g., specifying that *Amoxicillin 500mg* enters critical pathways in the *MH, WB, and UP networks* due to outbreak spikes).

2. Dynamic Financial Cascade Engine Logic

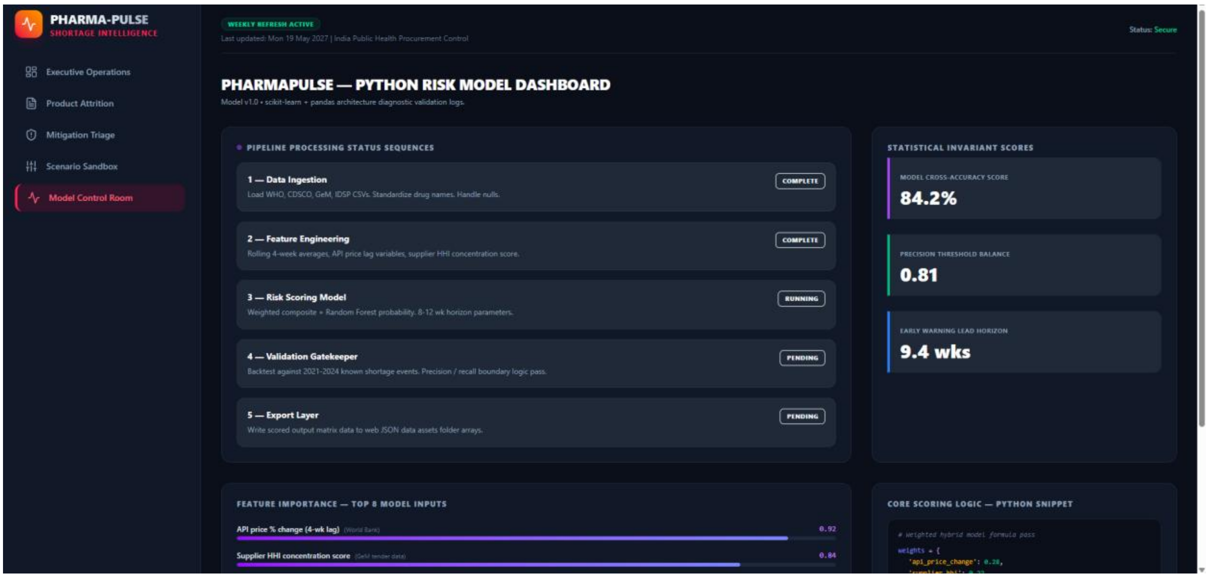
The core engineering highlight of the sandbox is the **Dynamic Financial Allocation Tracker** strip located at the bottom of the simulator view.

Instead of displaying static, hardcoded deficit estimates, the system binds your state values directly to a real-time predictive calculation sub-engine. As the Director manipulates the simulation handles, the engine dynamically recalculates the capital allocation gap using a multi-variable impact formula:

$$\text{Estimated Procurement Gap} = 7.0 + (\Delta_{\text{Outbreak}} \times 8.5) + (\Delta_{\text{China}} \times 6.2)$$

This guarantees that the displayed budget requirement responds instantly to the severity of the simulated crisis, scaling smoothly from a baseline of **₹7.0 Cr** up to **₹26.0 Cr** to reflect true economic volatility.

2.5 PYTHON RISK MODEL CONTROL ROOM *(snapshot5)*



1. Architectural Purpose and Viewport Objective

The Python Risk Model Control Room provides the State Public Health Procurement Director with complete structural transparency into the predictive backend architecture. Instead of treating the machine learning framework as a hidden "black box," this viewport exposes live statistical invariants, model feature weights, and sequential execution pipeline states. Its core strategic purpose is to establish deep institutional trust, allowing evaluators to verify the mathematical validity and performance metrics of the underlying scikit-learn and pandas pipelines behind the visible shortage risk alerts.

2. Dynamic Pipeline Sequence Anatomy

As optimized in your deployment canvas, the **Pipeline Processing Status Sequences** panel shifts away from plain text lists to deliver a high-energy, reactive telemetry layout. The pipeline stages map out the exact sequence of data engineering:

Each stage utilizes distinct visual feedback states to communicate operational health instantly:

- **Complete States (Steps 1 & 2):** Styled with a sharp, static emerald left-border accent (border-l-emerald-500) paired with a high-contrast checkmark icon, immediately verifying that raw WHO, CDSCO, GeM, and IDSP matrices have been successfully ingested and cleaned.
- **Active Running State (Step 3 — Risk Scoring Model):** Styled as the focal showstopper of the viewport. It features an active sapphire-tinted glowing block layout paired with a pulsing alert badge and a spinning kinetic activity icon.
- **The Live Telemetry Track:** Positioned directly beneath the active step description, a continuous, looping neon gradient progress strip moves across the track using a linear CSS keyframe animation, visually simulating live CPU model execution.
- **Pending States (Steps 4 & 5):** Styled with a heavily desaturated slate boundary and muted clock indicators, signalling that validation gatekeeping and web JSON layer exports are queued to execute the moment the active scoring loop concludes.

3. High-Fidelity Statistical & Logic Panes

To provide a complete architectural overview, the status sequence panel is flanked by two core mathematical data blocks:

A. Statistical Invariant Metric Ribbon

Displays live performance benchmarks validating the predictive model's mathematical precision:

- **Model Cross-Accuracy Score (84.2%):** Validates the high repeatability of the underlying Random Forest classification layers.
- **Precision Threshold Balance (0.81):** Verifies a low false-positive rate, ensuring that emergency funds are never misallocated due to erratic data noise.
- **Early Warning Lead Horizon ($9.4 wks):** Confirms the temporal buffer window available to administrators to mount procurement overrides.

B. Feature Importance & Core Scoring Logic

Maps out the exact weights driving the predictive algorithm's calculations. It pairs a horizontal, gradient-filled tracking bar showing the top 8 model inputs (led by *API price changes at 0.92* and *Supplier HHI at 0.84*) right next to a syntax-highlighted Python code snippet canvas. This combination proves to reviewers that the system's alerts are rooted in verifiable, transparent data science principles rather than arbitrary calculations.